

Practice #4: Synthesis with Genus.

Make a synthesis of the “full-adder” of the practice 2 using the basic library for genus to get the report of area, power, delay and the schematic of the circuit build.

Repeat the previous steps but this time using the complete library.

What are the differences between each report?

After that choose the one that the best performance and predict what do you expect for the reports of area, power and delay but this time for the adder of the practice #3 and comment if your thoughts was right.

Using the basic library:

“Area report”

```
@genus:root: 8> report area
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Sep 20 2024 02:01:59 am
Module:            Full_Adder
Technology library: \home\aroman\cadence\verilog_lib\LIBRARY\ami06_ccs 1.0
Operating conditions: PVT_5V_25C (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====

Instance  Module  Cell Count  Cell Area  Net Area  Total Area  Wireload
-----
Full_Adder      13    146.362    0.000    146.362 <none> (D)
(D) = wireload is default in technology library
@genus:root: 9> █
```

“Power report”

```

@genus:root: 10> report power
Info    : Joules engine is used. [RPT-16]
        : Joules engine is being used for the command report_power.
Info    : ACTP-0001 [ACTPInfo] Activity propagation started for stim#0 netlist
        : Full_Adder
Info    : ACTP-0009 [ACTPInfo] Activity Propagation Progress Report : 100%
Info    : ACTP-0001 Activity propagation ended for stim#0
Info    : PWRA-0001 [PwrInfo] compute_power effective options
        : -mode : vectorless
        : -skip_propagation : 1
        : -frequency_scaling_factor : 1.0
        : -use_clock_freq : stim
        : -stim :/stim#0
        : -fromGenus : 1
Info    : ACTP-0001 Timing initialization started
Info    : ACTP-0001 Timing initialization ended
Info    : PWRA-0002 [PwrInfo] Skipping activity propagation due to -skip_ap
        : option....
Warning: PWRA-0302 [PwrWarn] Frequency scaling is not applicable for vectorless
        : flow. Ignoring frequency scaling.
Warning: PWRA-0304 [PwrWarn] -stim option is not applicable with vectorless mode
        : of power analysis, ignored this option.
Info    : PWRA-0002 Started 'vectorless' power computation.
Info    : PWRA-0002 Finished power computation.
Info    : PWRA-0007 [PwrInfo] Completed successfully.
        : Info=6, Warn=2, Error=0, Fatal=0
Instance: /Full_Adder
Power Unit: W
PDB Frames: /stim#0/frame#0
-----
      Category      Leakage      Internal      Switching      Total      Row%
-----
      memory      0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
      register    0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
      latch       0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
      logic       2.29010e-11  2.46673e-05  5.52469e-05  7.99142e-05  100.00%
      bbox       0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
      clock      0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
      pad        0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
      pm         0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
-----
      Subtotal    2.29010e-11  2.46673e-05  5.52469e-05  7.99142e-05  100.00%
      Percentage  0.00%      30.87%      69.13%      100.00%  100.00%
-----
@genus:root: 11> █

```

“Delay report”

```
@genus:root: 11> report timing -unconstrained
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Sep 20 2024  02:04:38 am
Module:           Full Adder
Operating conditions: PVT_5V_25C (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

Path 1: UNCONSTRAINED
  Startpoint: (F) B
    Endpoint: (F) S

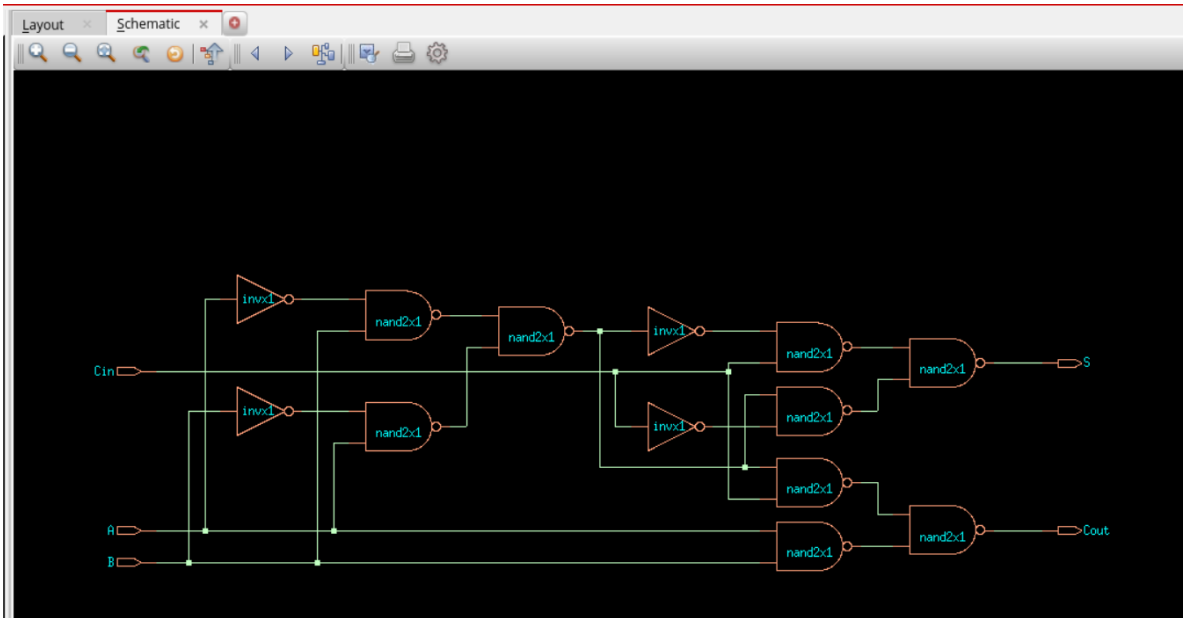
      Capture      Launch
Drv Adjust:++      0        0

      Data Path:-      723

#-----#
# Timing Point  Flags      Arc      Edge      Cell      Fanout Load Trans Delay Arrival Instance
#                                     (fF)  (ps)  (ps)  (ps)  Location
#-----#
B              -          -          F      (arrival)    3 27.1      0      0      0      (-,-)
g71/out        -          A->out R      invx1        1 10.0     44     109     109     (-,-)
g67__6783/out  -          A->out F      nand2x1       1 9.8      48     104     213     (-,-)
g66__5526/out  -          B->out R      nand2x1       3 27.5     109    175     389     (-,-)
g65/out        -          A->out F      invx1        1 10.0     69     149     538     (-,-)
g62__6260/out  -          A->out R      nand2x1       1 10.0     51     115     652     (-,-)
g60__2398/out  -          A->out F      nand2x1       1 0.0      23      70     723     (-,-)
S              -          -          F      (port)       -      -      -      0     723     (-,-)
#-----#

@genus:root: 12> █
```

“Squematic”



Using the complete library:

“Area report”

```
@genus:root: 8> report area
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Sep 20 2024  02:35:54 am
Module:            Full_Adder
Technology library: gsc145nm
Operating conditions: typical (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====

Instance  Module  Cell Count  Cell Area  Net Area  Total Area  Wireload
-----
Full_Adder          1      8.917    0.000      8.917 <none> (D)
(D) = wireload is default in technology library
@genus:root: 9> █
```

“Power report”

```

@genus:root: 10> report power
Info    : Joules engine is used. [RPT-16]
        : Joules engine is being used for the command report_power.
Info    : ACTP-0001 [ACTPInfo] Activity propagation started for stim#0 netlist
        : Full_Adder
Info    : ACTP-0009 [ACTPInfo] Activity Propagation Progress Report : 100%
Info    : ACTP-0001 Activity propagation ended for stim#0
Info    : PWRA-0001 [PwrInfo] compute_power effective options
        : -mode : vectorless
        : -skip_propagation : 1
        : -frequency_scaling_factor : 1.0
        : -use_clock_freq : stim
        : -stim :/stim#0
        : -fromGenus : 1
Info    : ACTP-0001 Timing initialization started
Info    : ACTP-0001 Timing initialization ended
Info    : PWRA-0002 [PwrInfo] Skipping activity propagation due to -skip_ap
        : option....
Warning: PWRA-0302 [PwrWarn] Frequency scaling is not applicable for vectorless
        : flow. Ignoring frequency scaling.
Warning: PWRA-0304 [PwrWarn] -stim option is not applicable with vectorless mode
        : of power analysis, ignored this option.
Info    : PWRA-0002 Started 'vectorless' power computation.
Info    : PWRA-0002 Finished power computation.
Info    : PWRA-0007 [PwrInfo] Completed successfully.
        : Info=6, Warn=2, Error=0, Fatal=0
Instance: /Full_Adder
Power Unit: W
PDB Frames: /stim#0/frame#0
-----
Category      Leakage      Internal      Switching      Total      Row%
-----
memory        0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
register       0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
latch         0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
logic         3.92104e-08  3.65938e-07  3.50900e-07  7.56049e-07  100.00%
bbox          0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
clock         0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
pad           0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
pm            0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
-----
Subtotal      3.92104e-08  3.65938e-07  3.50900e-07  7.56049e-07  100.00%
Percentage    5.19%      48.40%      46.41%      100.00%  100.00%
-----
@genus:root: 11> 

```

“Delay report”

```
@genus:root: 11> report timing -unconstrained
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Sep 20 2024  02:38:24 am
Module:           Full_Adder
Operating conditions: typical (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

Path 1: UNCONSTRAINED
Startpoint: (R) B
Endpoint: (F) S

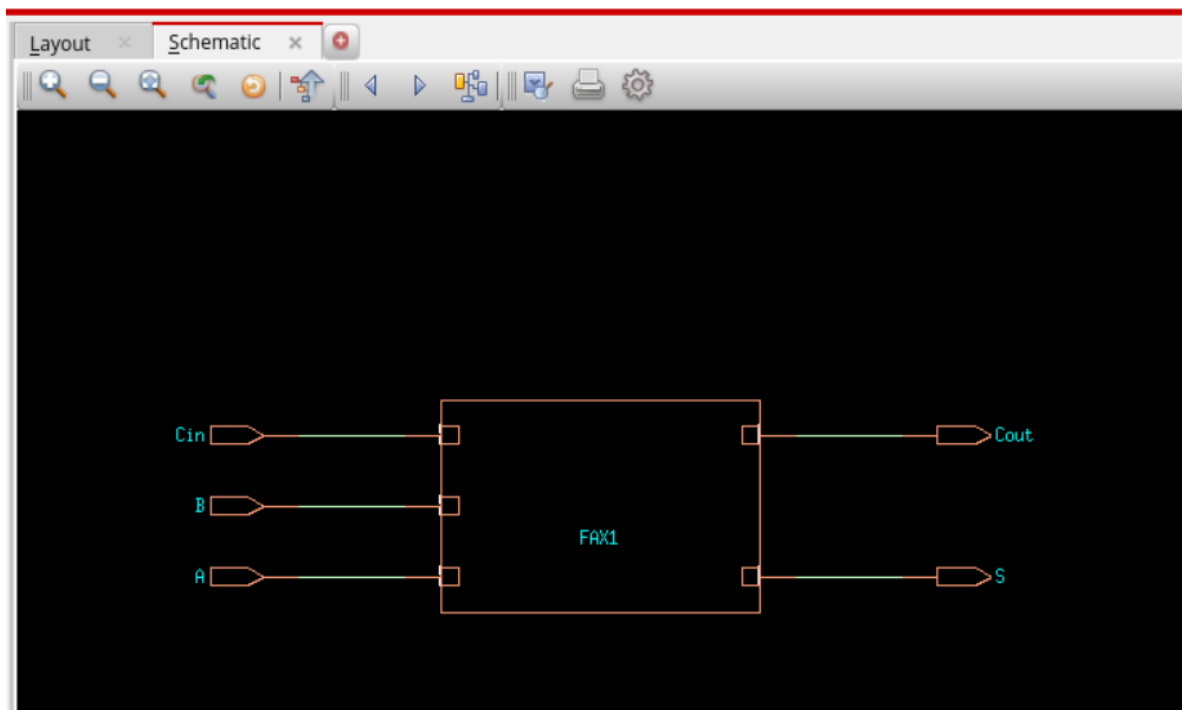
      Capture      Launch
Drv Adjust: +      0      0

      Data Path: -      77

#-----#
# Timing Point  Flags  Arc  Edge  Cell      Fanout Load Trans Delay Arrival Instance
#              (fF)  (ps)  (ps)  (ps)  (ps)  Location
#-----#
#   B              -    -    R    (arrival)    1 10.3      0    0      0    (-,-)
#   g2/YS          -    B->YS F    FAX1         1 0.0      3    77     77    (-,-)
#   S              -    -    F    (port)         -  -        -    0     77    (-,-)
#-----#

@genus:root: 12> █
```

Schematic



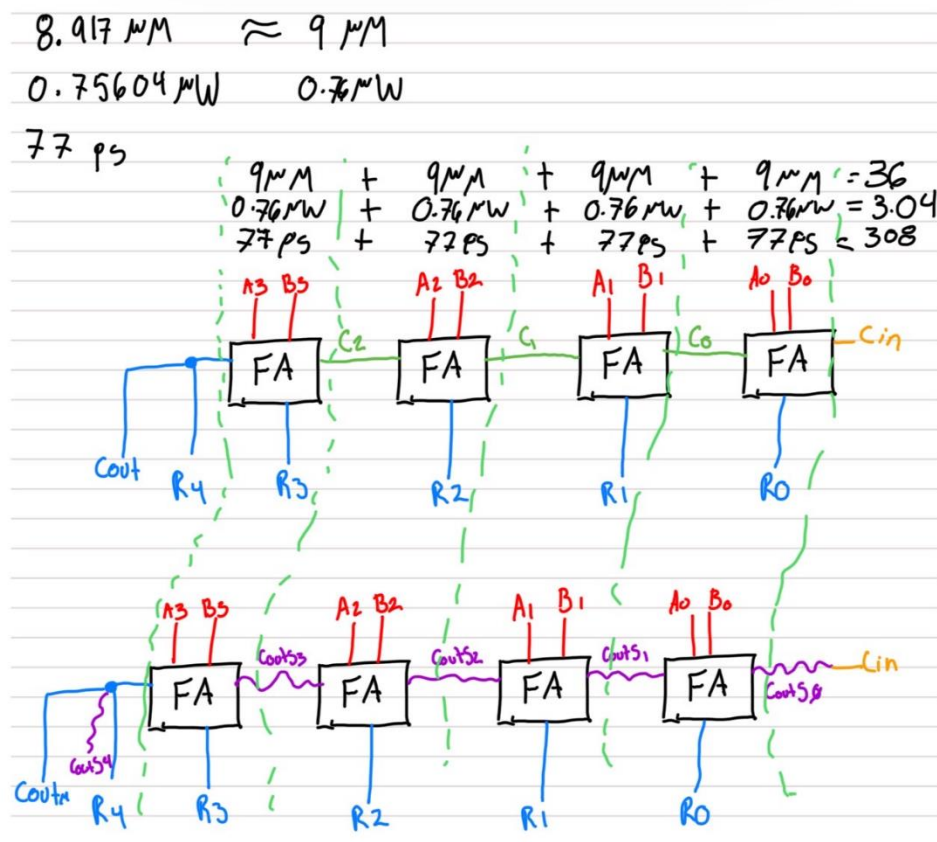
After seeing both results, we have:

	Basic Library	Complete Library	Units
Area	146.362	8.917	uM
Power	79.9142	0.75604	uW
Delay	723	77	ps

We see that there is a huge improvement using the complete library this is because we have much more components that can do a single chore and with the basic library well we need to use a mix of the gates so they can make the same job than one but with the complete library.

Even in the schematic we can see how the basic library uses a lot of gates to do the full-adder but in the case of the complete library there is all ready a component of a full adder.

With this logic and using the complete library and if we know that the adder of the practice 3 is a carry ripple adder that basically we can say that it is 4 full-adders on series.



The area will be around 27 uM.

The power will be around 2.28 uW

The delay will be around 231ps

And just for practice we have 2 examples of 4bits full adder one with generate and one using structural description.

Adder with structural description:

“Area report”

```
@genus:root: 8> report area
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Sep 21 2024  01:52:08 am
Module:           Full_Adder_4b
Technology library: gsc145nm
Operating conditions: typical (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

  Instance  Module  Cell Count  Cell Area  Net Area  Total Area  Wireload
-----
Full_Adder_4b      4      35.667    0.000      35.667 <none> (D)
(D) = wireload is default in technology library
@genus:root: 9> █
```

“Power report”

```

@genus:root: 10> report power
Info    : Joules engine is used. [RPT-16]
         : Joules engine is being used for the command report_power.
Info    : ACTP-0001 [ACTPInfo] Activity propagation started for stim#0 netlist
         : Full_Adder_4b
Info    : ACTP-0009 [ACTPInfo] Activity Propagation Progress Report : 100%
Info    : ACTP-0001 Activity propagation ended for stim#0
Info    : PWRA-0001 [PwrInfo] compute_power effective options
         : -mode : vectorless
         : -skip_propagation : 1
         : -frequency_scaling_factor : 1.0
         : -use_clock_freq : stim
         : -stim :/stim#0
         : -fromGenus : 1
Info    : ACTP-0001 Timing initialization started
Info    : ACTP-0001 Timing initialization ended
Info    : PWRA-0002 [PwrInfo] Skipping activity propagation due to -skip_ap
         : option....
Warning: PWRA-0302 [PwrWarn] Frequency scaling is not applicable for vectorless
         : flow. Ignoring frequency scaling.
Warning: PWRA-0304 [PwrWarn] -stim option is not applicable with vectorless mode
         : of power analysis, ignored this option.
Info    : PWRA-0002 Started 'vectorless' power computation.
Info    : PWRA-0002 Finished power computation.
Info    : PWRA-0007 [PwrInfo] Completed successfully.
         : Info=6, Warn=2, Error=0, Fatal=0
Instance: /Full_Adder_4b
Power Unit: W
PDB Frames: /stim#0/frame#0
-----
Category      Leakage      Internal      Switching      Total      Row%
-----
memory        0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
register      0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
latch        0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
logic        1.56842e-07  1.99531e-06  1.55016e-06  3.70232e-06  100.00%
bbox         0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
clock        0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
pad          0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
pm           0.000000e+00  0.000000e+00  0.000000e+00  0.000000e+00  0.00%
-----
Subtotal     1.56842e-07  1.99531e-06  1.55016e-06  3.70232e-06  100.00%
Percentage   4.24%      53.89%      41.87%      100.00%  100.00%
-----
@genus:root: 11> 

```

“Delay report”

```

@genus:root: 11> report timing -unconstrained
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Sep 21 2024  01:54:36 am
Module:            Full_Adder_4b
Operating conditions:  typical (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====

Path 1: UNCONSTRAINED
  Startpoint: (F) CinM
    Endpoint: (F) Result[3]

      Drv Adjust:+      Capture      Launch
      Data Path:-      0              0

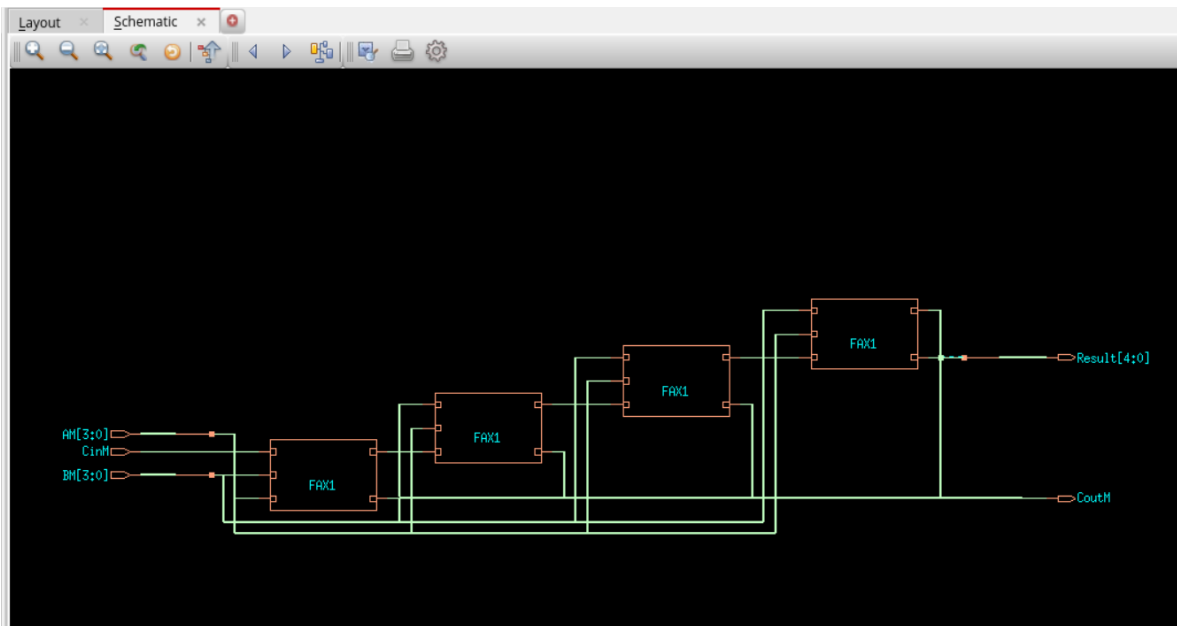
      Data Path:-      322

#-----#
# Timing Point  Flags  Arc  Edge  Cell      Fanout Load Trans Delay Arrival Instance
#                                     (fF)  (ps)  (ps)  (ps)  Location
#-----#
CinM              -      -    F    (arrival)    1 10.9      0      0      0      (-,-)
g233/YC           -      A->YC F    FAX1         1  7.1     29     84     84      (-,-)
g232/YC           -      C->YC F    FAX1         1  7.1     29     82    166      (-,-)
g231/YC           -      C->YC F    FAX1         1  7.1     29     82    248      (-,-)
g2/YS             -      C->YS F    FAX1         1  0.0       3     74    322      (-,-)
Result[3]         -      -    F    (port)       -   -       -      0    322      (-,-)
#-----#

@genus:root: 12>

```

Schematic



Adder with generate;

“Area report”

```
@genus:root: 8> report area
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Sep 21 2024 02:02:01 am
Module:            Full_Adder_generate
Technology library: gscl45nm
Operating conditions: typical (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====

  Instance      Module  Cell Count  Cell Area  Net Area  Total Area  Wireload
-----
Full_Adder_generate      4      35.667      0.000      35.667 <none> (D)
(D) = wireload is default in technology library
@genus:root: 9> █
```

“Power report”

```

@genus:root: 10> report power
Info      : Joules engine is used. [RPT-16]
           : Joules engine is being used for the command report_power.
Info      : ACTP-0001 [ACTPInfo] Activity propagation started for stim#0 netlist
           : Full_Adder_generate
Info      : ACTP-0009 [ACTPInfo] Activity Propagation Progress Report : 100%
Info      : ACTP-0001 Activity propagation ended for stim#0
Info      : PWRA-0001 [PwrInfo] compute_power effective options
           : -mode : vectorless
           : -skip_propagation : 1
           : -frequency_scaling_factor : 1.0
           : -use_clock_freq : stim
           : -stim :/stim#0
           : -fromGenus : 1
Info      : ACTP-0001 Timing initialization started
Info      : ACTP-0001 Timing initialization ended
Info      : PWRA-0002 [PwrInfo] Skipping activity propagation due to -skip_ap
           : option....
Warning: PWRA-0302 [PwrWarn] Frequency scaling is not applicable for vectorless
           : flow. Ignoring frequency scaling.
Warning: PWRA-0304 [PwrWarn] -stim option is not applicable with vectorless mode
           : of power analysis, ignored this option.
Info      : PWRA-0002 Started 'vectorless' power computation.
Info      : PWRA-0002 Finished power computation.
Info      : PWRA-0007 [PwrInfo] Completed successfully.
           : Info=6, Warn=2, Error=0, Fatal=0
Instance: /Full_Adder_generate
Power Unit: W
PDB Frames: /stim#0/frame#0
-----
Category      Leakage      Internal      Switching      Total      Row%
-----
memory        0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
register       0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
latch         0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
logic         1.56842e-07  1.99531e-06  1.55016e-06  3.70232e-06  100.00%
bbox          0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
clock         0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
pad           0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
pm            0.00000e+00  0.00000e+00  0.00000e+00  0.00000e+00  0.00%
-----
Subtotal      1.56842e-07  1.99531e-06  1.55016e-06  3.70232e-06  100.00%
Percentage    4.24%      53.89%      41.87%      100.00%  100.00%
-----
@genus:root: 11> █

```

“Delay report”

```

@genus:root: 11> report timing -unconstrained
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Sep 21 2024  02:03:36 am
Module:            Full_Adder_generate
Operating conditions:  typical (balanced_tree)
Wireload mode:     enclosed
Area mode:         timing library
=====

Path 1: UNCONSTRAINED
  Startpoint: (F) CinM
  Endpoint: (F) Result[3]

      Drv Adjust: +      Capture      Launch
                        0              0

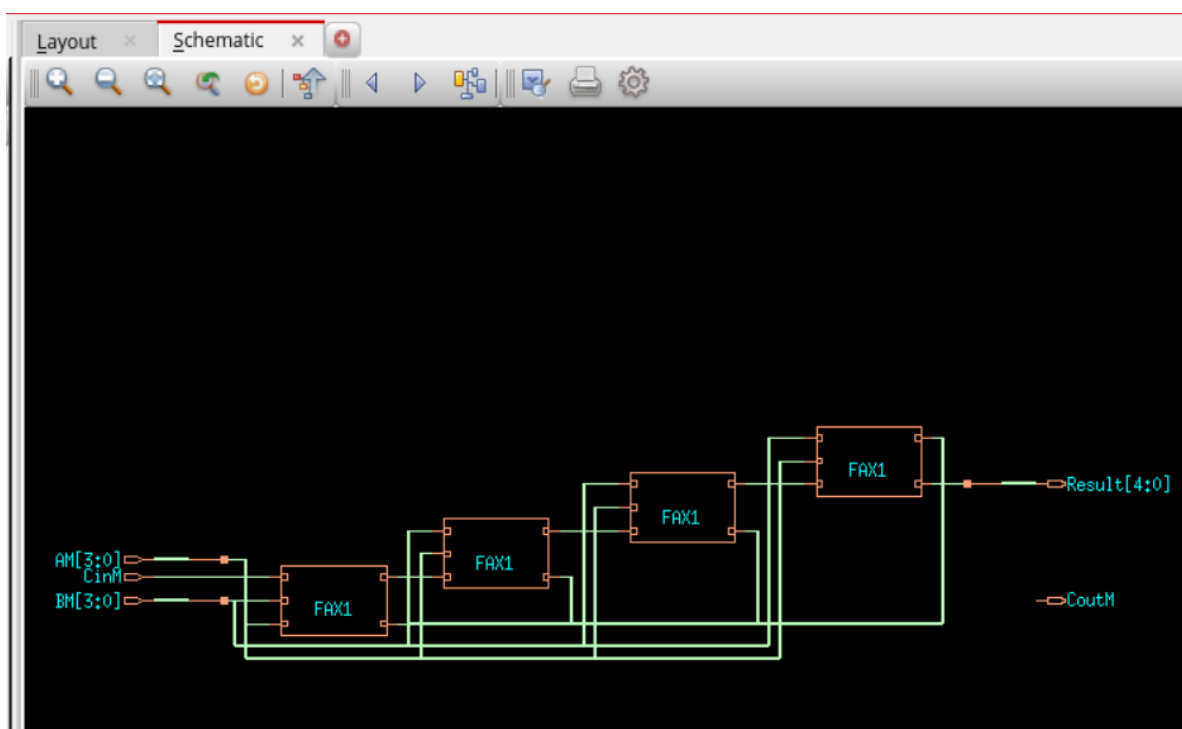
      Data Path: -      322

#-----#
# Timing Point  Flags   Arc   Edge   Cell      Fanout Load Trans Delay Arrival Instance
#              (fF)   (ps)   (ps)   (ps)   (ps)   (ps)   (ps)   (ps)   Location
#-----#
CinM           -       -     F   (arrival)    1 10.9      0      0      0      0      (-,-)
g233/YC        -       A->YC F   FAX1        1  7.1     29     84     84     84      (-,-)
g232/YC        -       C->YC F   FAX1        1  7.1     29     82    166    166      (-,-)
g231/YC        -       C->YC F   FAX1        1  7.1     29     82    248    248      (-,-)
g2/YS          -       C->YS F   FAX1        1  0.0       3     74    322    322      (-,-)
Result[3]      -       -     F   (port)       -    -      -      0    322    322      (-,-)
#-----#

@genus:root: 12> █

```

Schematic



	Prediction	Estructural Adder	Generate Adder	Units
Area	36	35.667	35.667	uM
Power	3.04	3.70232	3.70232	uW
Delay	308	322	322	ps

I can say that we are very close to the predictions obviously we need to understand that considering fan out, sizing, etc. the results can be different. And also, we can see that genus interpretate both adders as the same way.